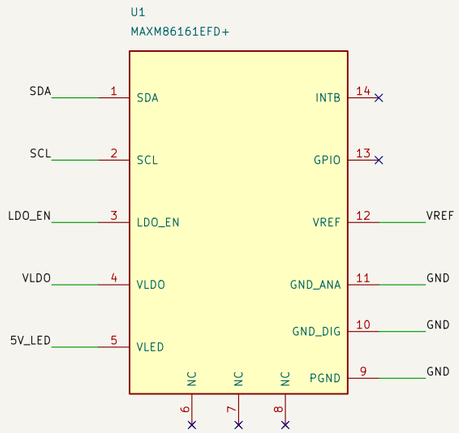
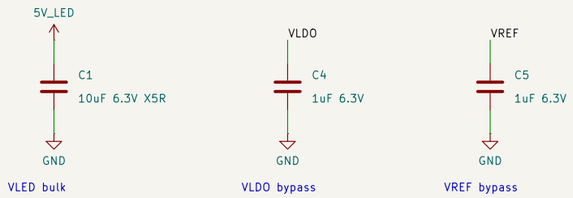
# SleepBud | PPG sensor board

v0.8 / MAXM86161 + STM32C011 / full-duplex UART over RS-422

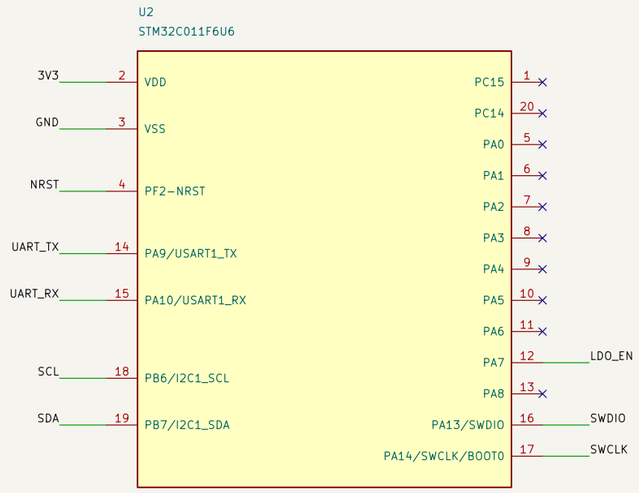
A3 | FUNCTIONAL SCHEMATIC

## 01 PPG SENSOR




## 02 MCU + I2C PULL-UPS

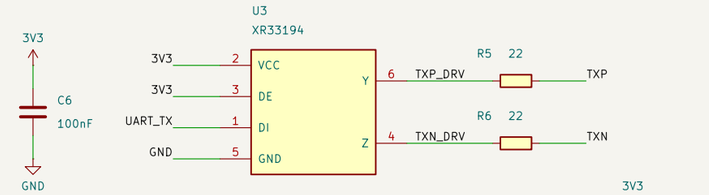






## 03 UART / RS-422 LINK

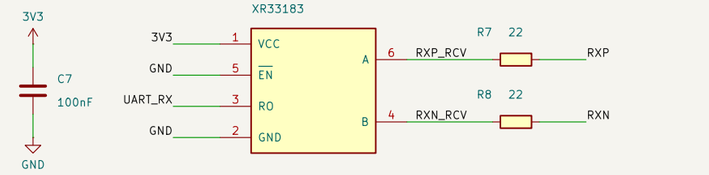
TX / earbud to controller

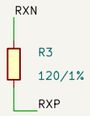






RX / controller to earbud



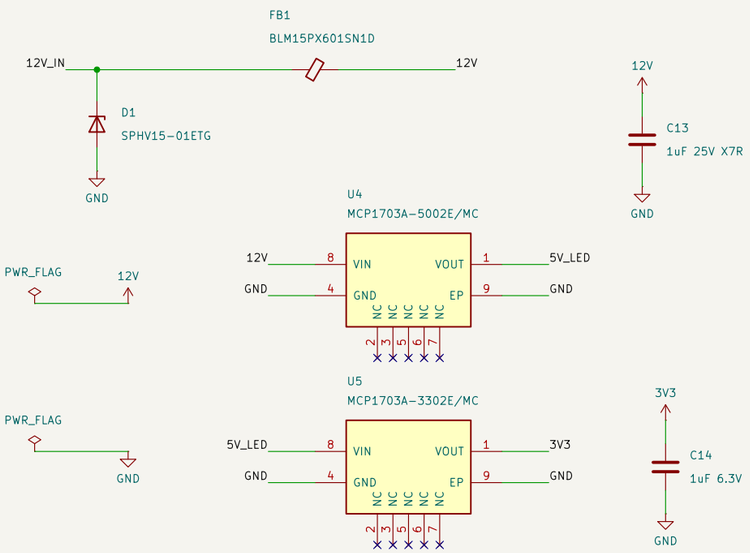




U3: DE = 3V3. U6: /EN = GND.

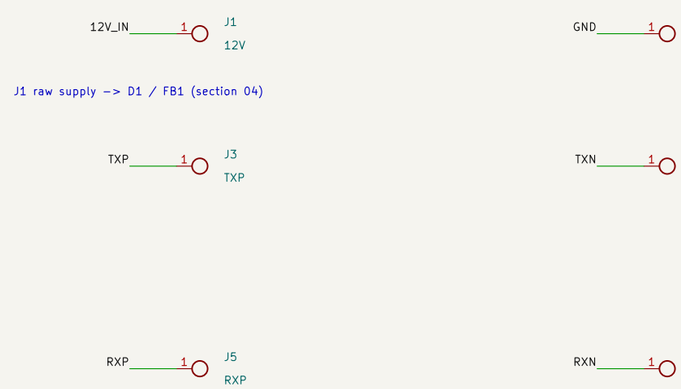
R3: 120/1% at the local RX cable pair.

Enhanced fail-safe: UART_RX high.

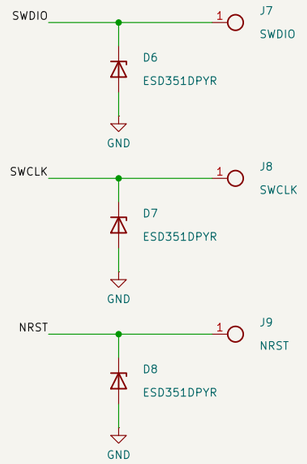
## 04 INPUT PROTECTION + POWER




## 05 RS-422 CABLE PADS




## 06 SWD TEST PADS + ESD




C1: 5V_LED / U4 output bulk capacitor (section 01). D1 clamps raw 12V_IN; FB1 feeds filtered 12V.

Cable: 12V, GND, TXP, TXN, RXP, RXN. Piezo coax bypasses this PCB. Matching net labels are connected.

RS-422 bus TVS omitted; 22 ohm series retained. D1 protects supply. D6-D8 protect SWD. IEC immunity requires physical validation.

PPG sensor board for in-ear sleep device

**SleepBud**

Sheet: /

File: ppg_pcb_v08.kicad_sch

***Title: SleepBud PPG PCB v0.8***

| Size: A3 | Date: | **Rev: 0.8** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 1/1 |